



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

SE 423: Introduction to Mechatronics

Lecture 10: UART, SPI, I2C

Marius Juston

Monday, February 23rd 2026

Office Hours:

- **Monday:** 4-6 PM, Neel
- **Tuesday:** 11-1 PM, Lakshmi
- **Tuesday:** 1-3 PM, Samuel
- **Tuesday:** 2-5 PM, Dan & Marius

Lab: Starting Lab 4 this week. This a 2-week lab!

Homeworks:

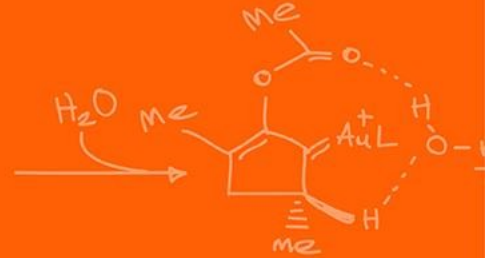
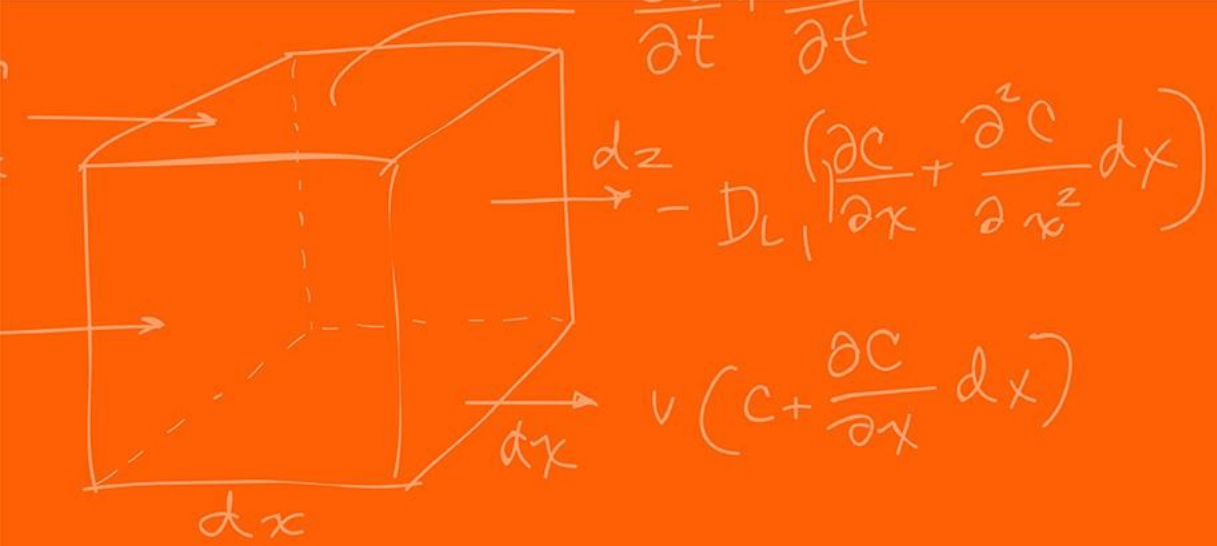
- Check-offs for [homework 2](#) Ex 3 & 4 are due **February 24, 5PM (The end of office hours)**
- Answers and code submissions for homework 2 are due **February 17, 9AM**

LabVIEW:

- All check-offs for [LabVIEW 2](#) are due **February 26, 5PM**

Questions?

Homework, Lab, LabVIEW questions?



Serial Communication



Communication between different signal could happen in 2 ways,

Parallel communication vs Serial communication

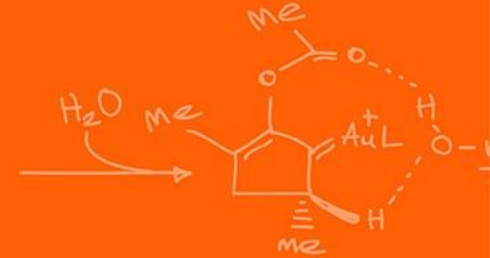
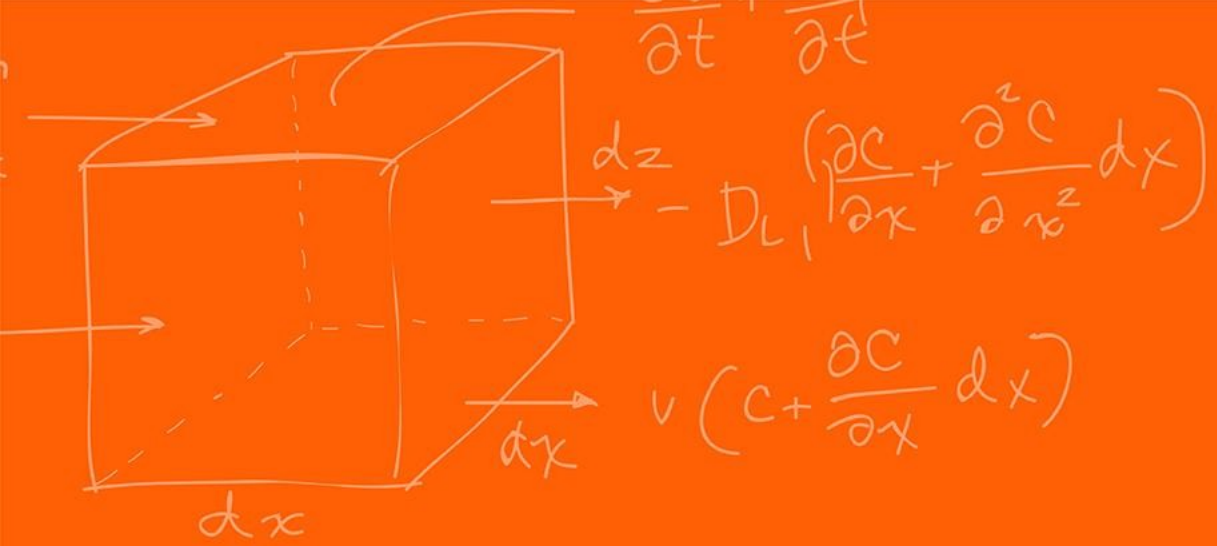
Why not do parallel communication?

- Requires many I/O pins, these are already very limited
- With the wires, we can run into electromagnetic interference (EMI) and cross-talk

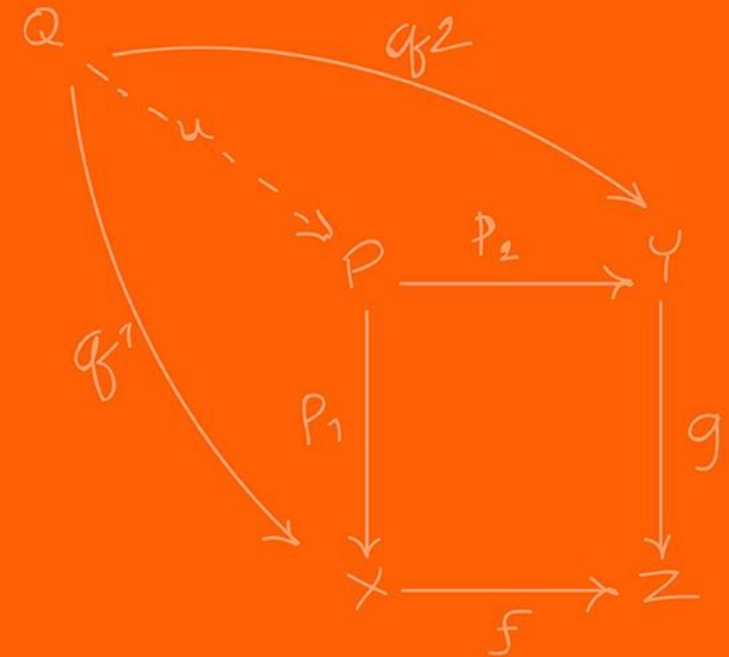
Serial communication, serializes data into single-bit stream, requires much less circuit complexity. What is the trade-off?

Time!

Serializing an N-bit word requires N discrete clock cycles!

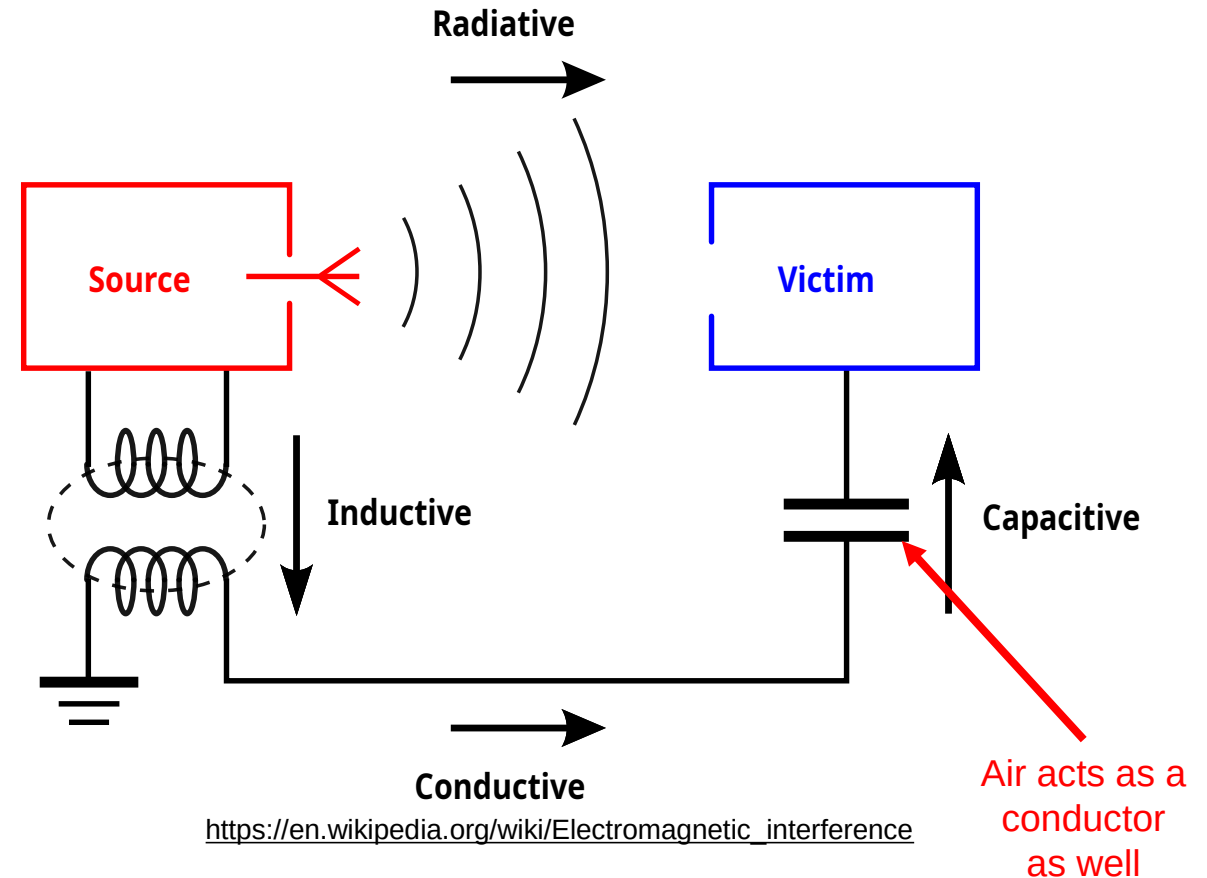


Interferences



There are 4 interference couplings:

- **Conductive:** You have physical connection causing interference
- **Inductive:** source and victim are separated by a short distance (typically less than a wavelength). Can be of two kinds, electrical induction and magnetic induction.
 - It is common to refer to electrical induction as **capacitive** coupling, and to magnetic induction as **inductive** coupling.
- **Capacitive:** occurs when a varying electrical field exists between two adjacent conductors, inducing a change in voltage on the receiving conductor.
- **Inductive:** magnetic coupling occurs when a varying magnetic field exists between two parallel conductors, inducing a change in voltage along the receiving conductor.
- **Radiative:** electromagnetic coupling occurs when source and victim are separated by a large distance. Source and victim act as radio antennas: the source emits or radiates an electromagnetic wave which is picked up or received by the victim.



Let's try to explain how we get capacitive interference!

Let $V_1(t)$ be the time-varying voltage on the noisy source wires and $V_2(t)$ be the voltage on the victim wire. Let C_{12} be the mutual stray capacitance between the two wires (could be air), let the victim wire have a resistance of R .

The fundamental equation for current through a capacitor is:

$$I = C \frac{dV}{dt}$$

The current injected from wire 1 into wire 2, I_{noise} , is driven from the difference between them across the capacitor.

$$I_{\text{noise}} = C_{12} \frac{d(V_1 - V_2)}{dt}$$

In practice, the noise source voltage is much larger than the victim's wire ($V_1 \gg V_2$), which allows the voltage derivative to simplify to:

$$I_{\text{noise}} = C_{12} \frac{d(V_1 - V_2)}{dt} \approx C_{12} \frac{dV_1}{dt}$$

By Ohm's law, the injected noise current flows through the victim's wire impedance to ground (R), creating a noise voltage:

$$V_{\text{noise}} \approx I_{\text{noise}} R \approx RC_{12} \frac{dV_1}{dt}$$

The interference voltage V_{noise} is directly proportional to the rate of change of the source voltage, even if the actual voltage amplitude is moderate!

What causes more digital signal interference?

High-frequency vs low-frequency

Assuming a duty cycle of 50%?

EPwm7Regs.TBPRD = 5000; EPwm7Regs.TBPRD = 500;

Let's try to explain how we get inductive interference!

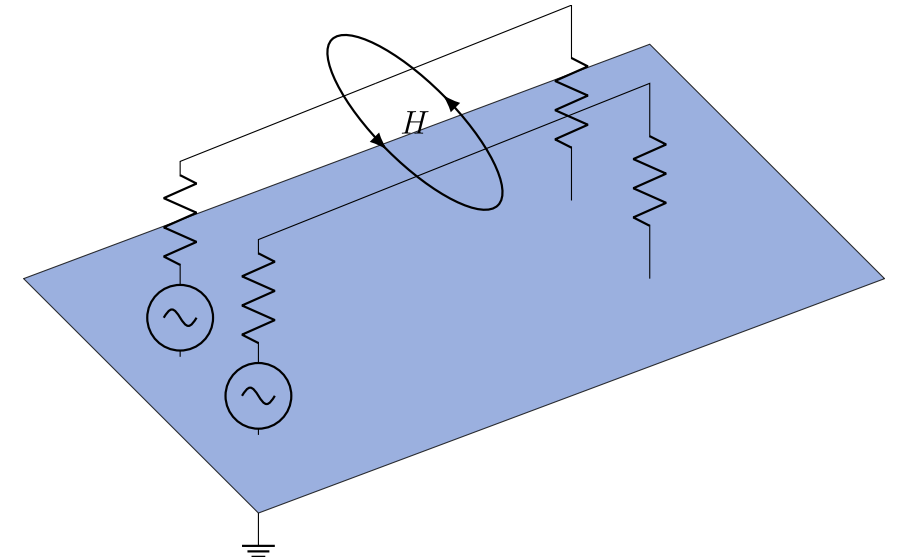
Coupling between the circuits can occur when the magnetic field lines from one of the circuits pass through the loop formed by the other circuit.

Magnetic coupling is governed by **Ampere's Law** and **Faraday's Law** of Induction. A time-varying current in a source wire generates a time-varying magnetic field, which penetrates the loop area formed by the victim wire and its ground return path, inducing an electromotive force (EMF).

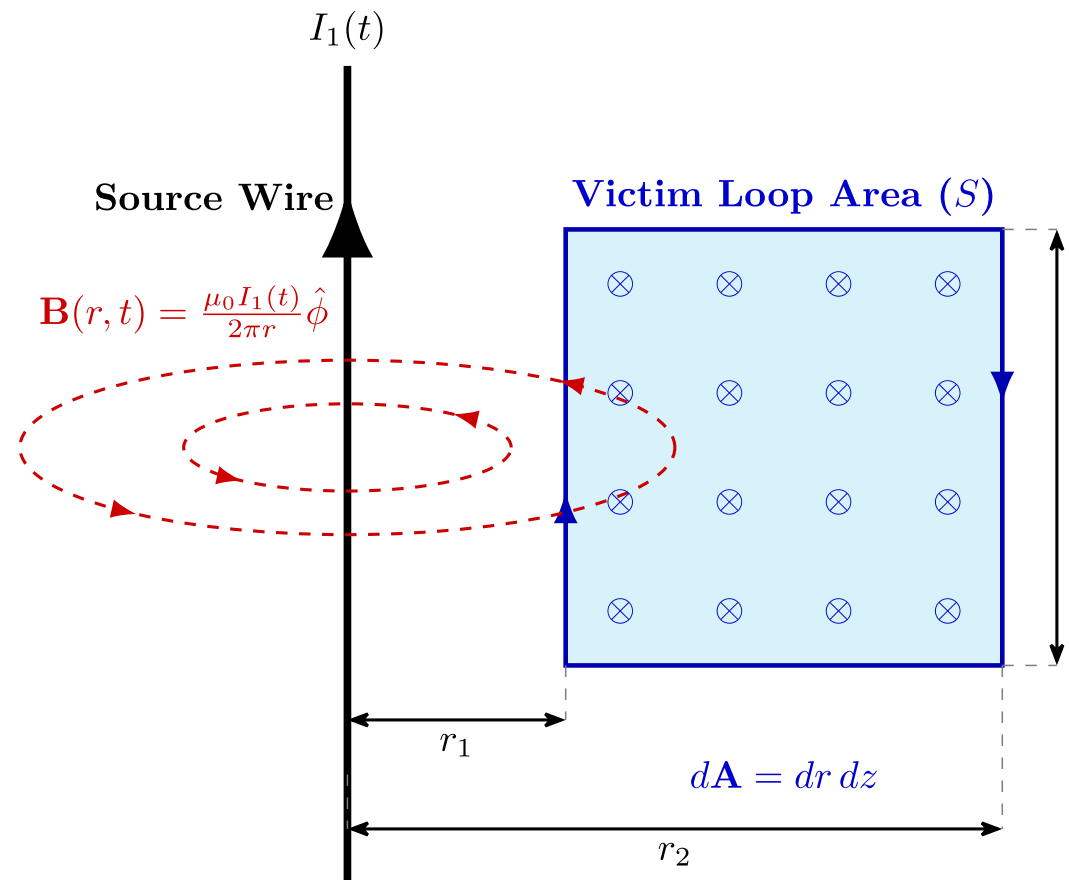
Let $I_1(t)$ be the time-varying current for. Assuming it is a long, straight wire, we use Ampere's Law ($\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$) to find the magnetic field at radial distance r :

$$B(r, t) = \frac{\mu_0 I_1(t)}{2\pi r} \hat{\phi}$$

(there μ_0 , is the permeability of free space and $\hat{\phi}$ represents the flux)



If the victim wire forms a rectangular loop (not often the case but good enough for showing this specific case), with its return path. Having a length l and distance r_1 and r_2 from the source wire. The magnetic flux Φ_B , though this loop is the surface integral of the magnetic field.



Animate this!

Integrating the field across the area.

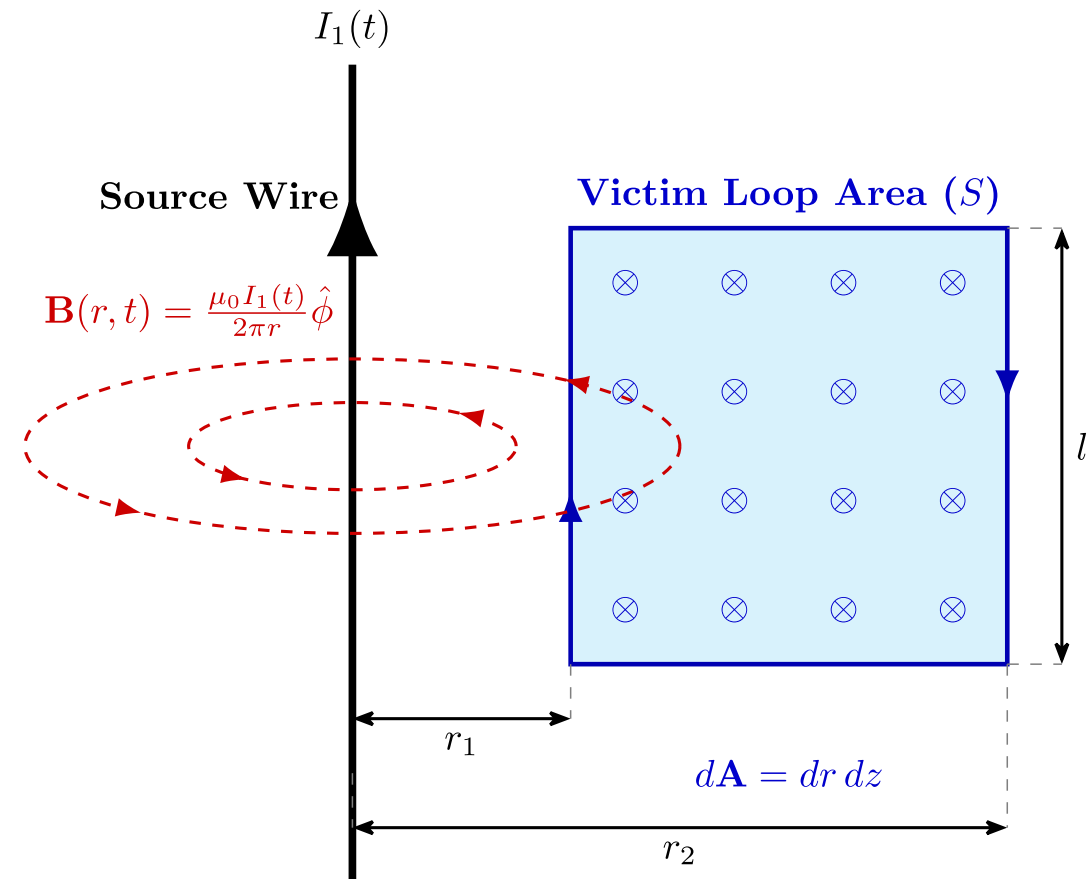
$$\Phi_B = \iint_S \mathbf{B} \cdot d\mathbf{A}$$

Substituting and integrating over the area $dA = l dr$:

$$\Phi_B = \int_{r_1}^{r_2} \left(\frac{\mu_0 I_1(t)}{2\pi r} \right) l dr$$

$$\Phi_B = \frac{\mu_0 l I_1(t)}{2\pi} \int_{r_1}^{r_2} \frac{1}{r} dr$$

$$\Phi_B = \frac{\mu_0 l I_1(t)}{2\pi} \ln \frac{r_2}{r_1}$$

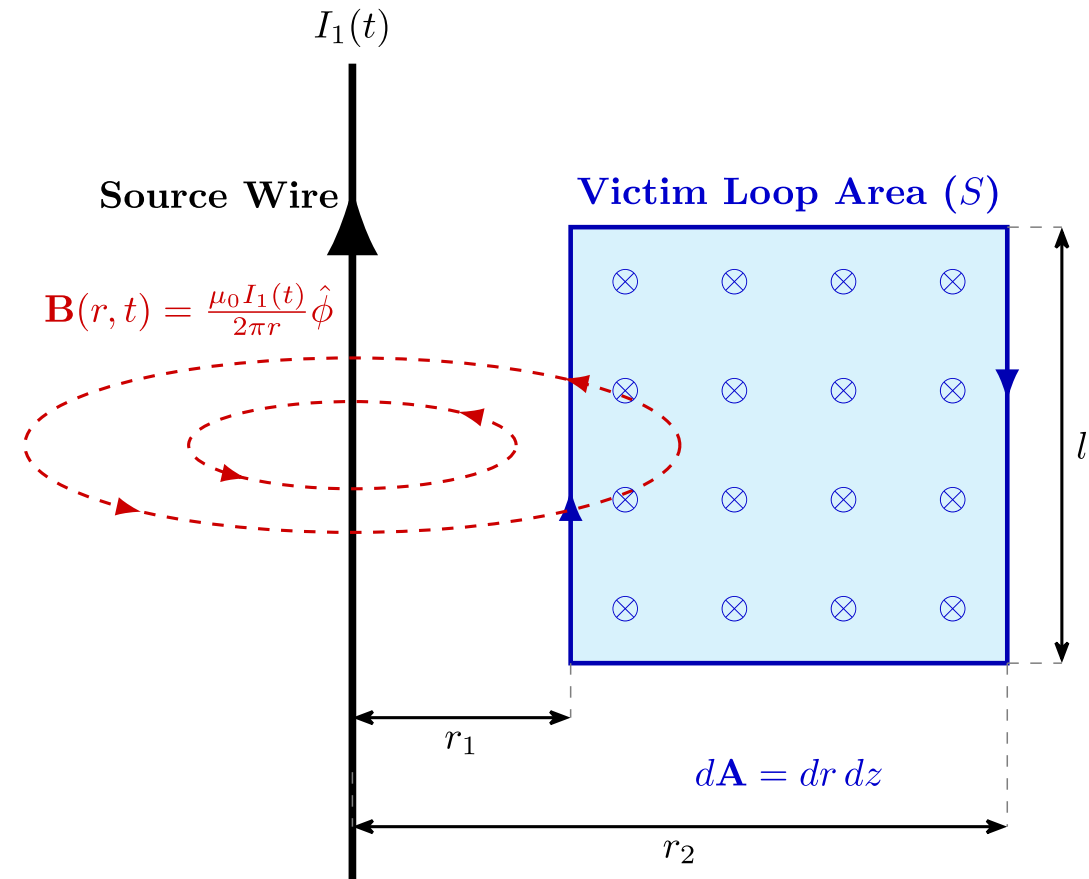


Faraday's Law states that the induced voltage (EMF) in a closed loop is equal to the negative rate of change of magnetic flux through the loop:

$$V_{\text{noise}} = -\frac{d\Phi_B}{dt}$$

Substituting we get:

$$V_{\text{noise}} = -\frac{\mu_0 l}{2\pi r} \ln \frac{r_2}{r_1} \times \frac{dI_1(t)}{dt}$$

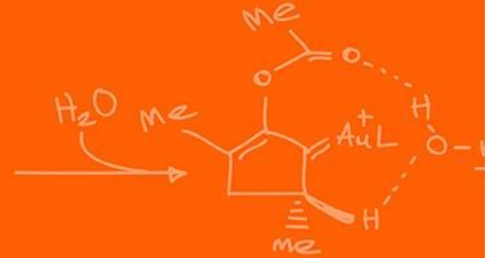
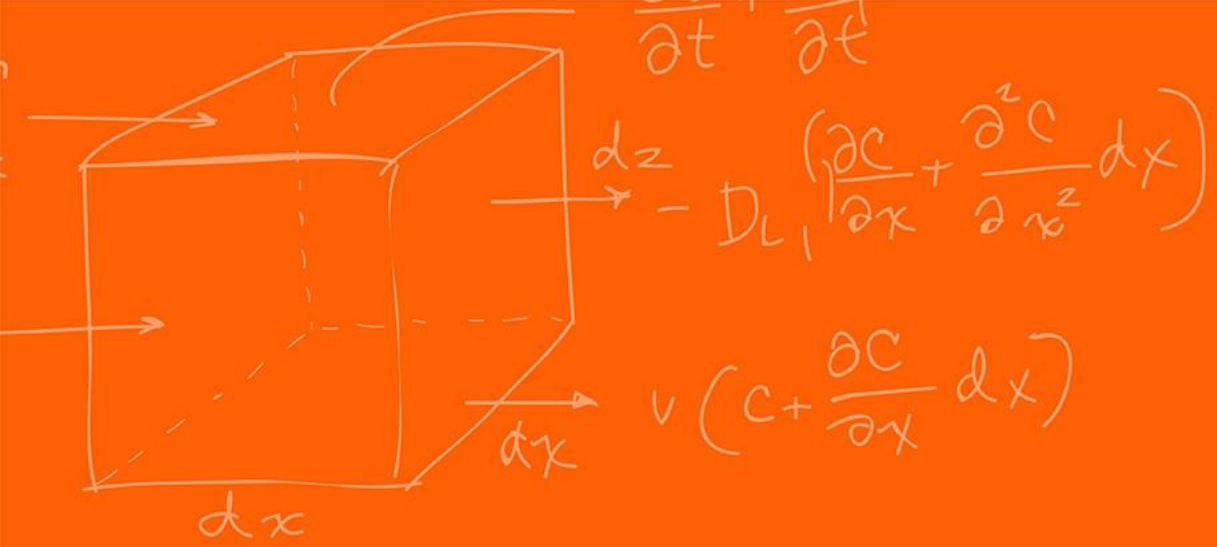


Thus, given the final interference noise,

$$V_{\text{noise}} = -\frac{\mu_0 l}{2\pi r} \ln \frac{r_2}{r_1} \times \frac{dI_1(t)}{dt}$$

The interference voltage V_{noise} is directly proportion to the rate of change of the current in the interfering wire!

This is why heavy machinery or circuits that draw massive, pulsating currents (like motor drives) are notorious sources of magnetic Electromagnetic Interference (EMI)!



Serial Communication



We want to be able to communicate with multiple devices. We can use a “Master / Slave” paradigm in communication.

“Master / Slave” is still commonly used but sometimes called Peripheral In and Peripheral Out.

The “Master” Node, holds the authority to initiate data transfers and generates the synchronization clock.

The “Slave” Nodes are strictly reactive and remain low until explicitly addressed.

Think of it as the “Master” node as a “Teacher” asking you to do something, the “Slave” nodes are the “students” doing the work and reporting the results back.

Communication between different signal could happen in 2 ways,

Parallel communication vs Serial communication

Why not do parallel communication?

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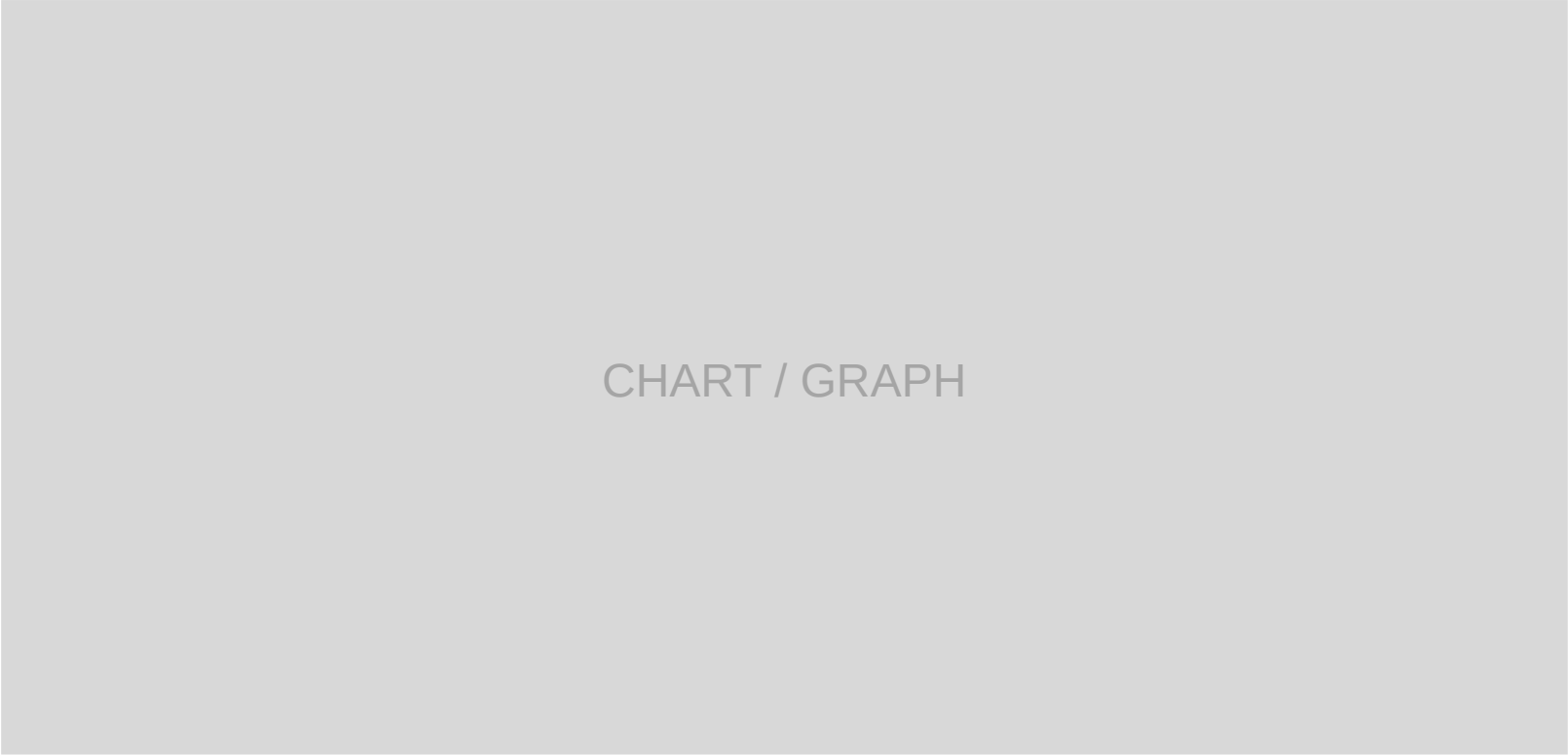
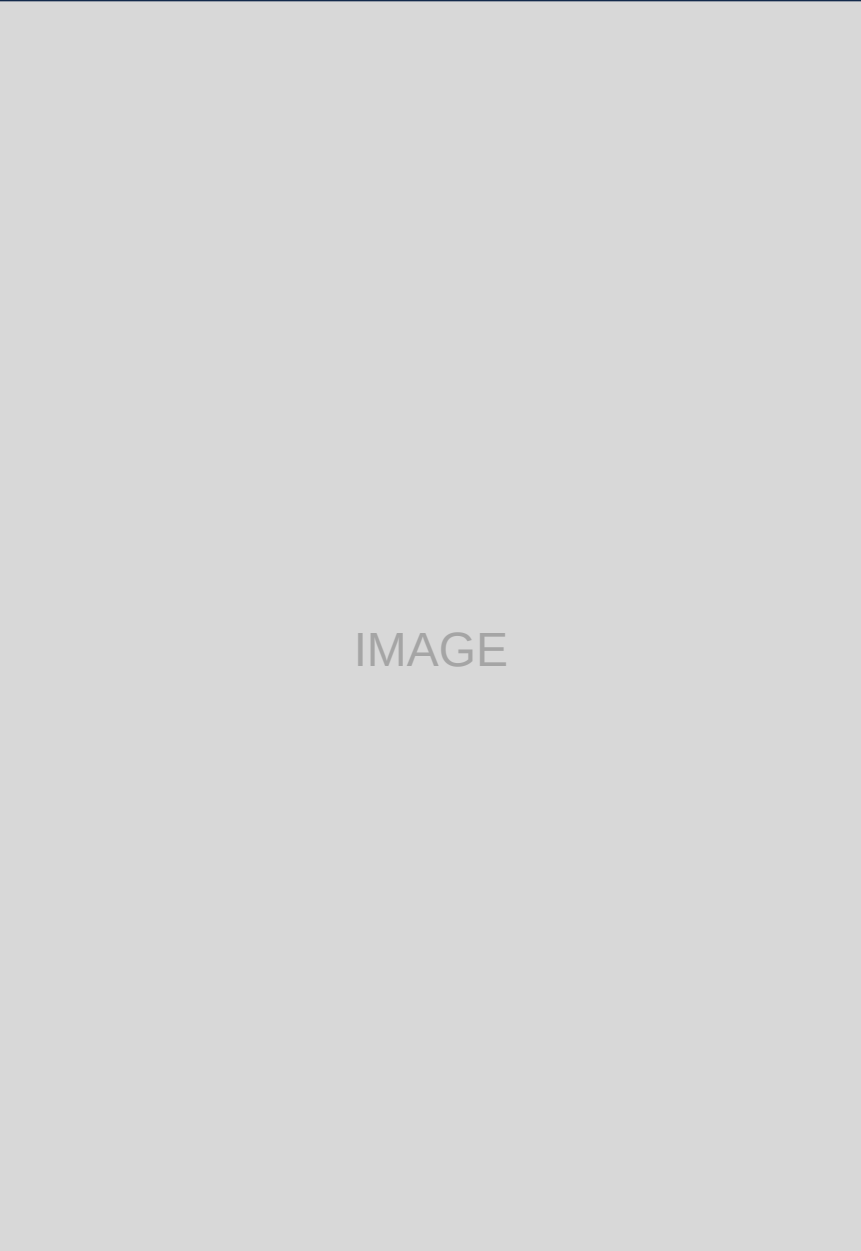


CHART / GRAPH

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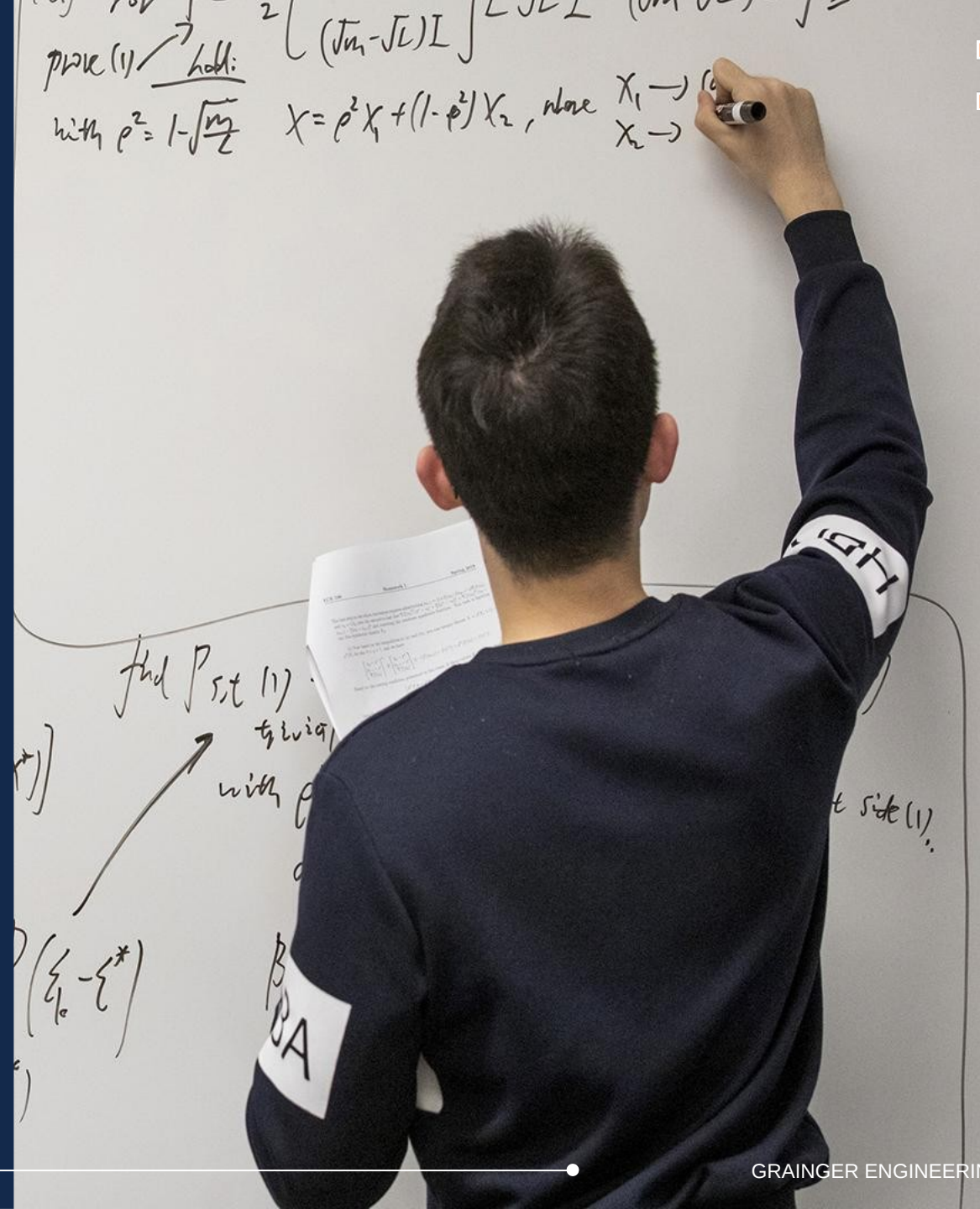
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IMAGE

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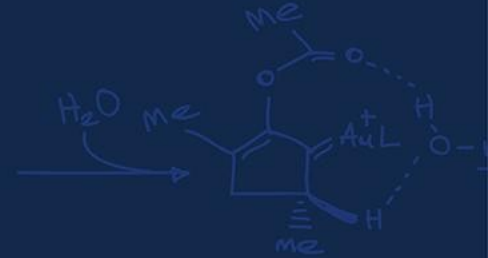
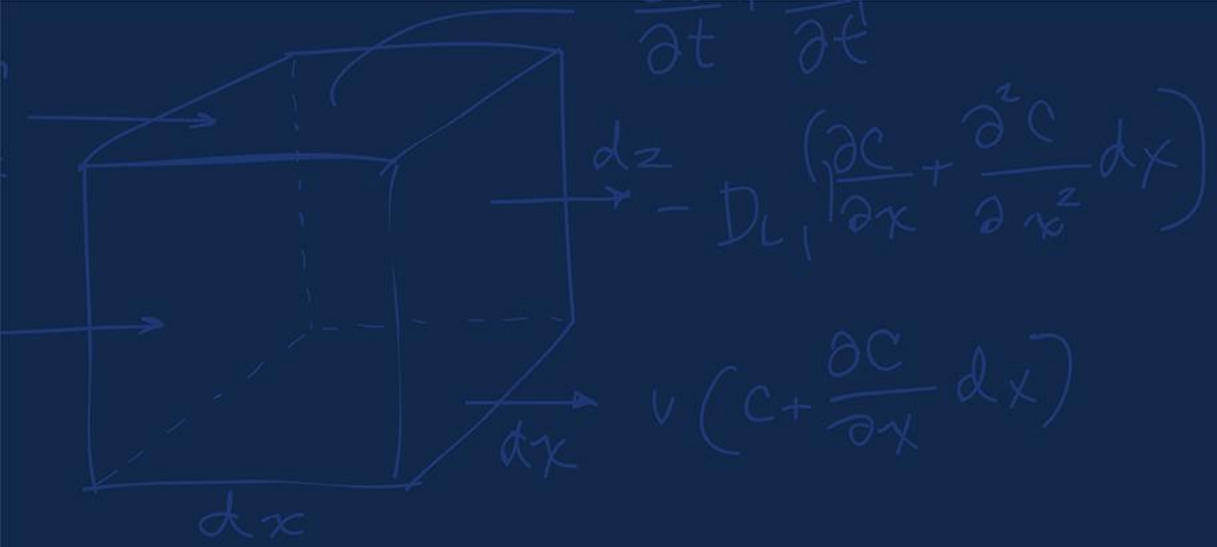
Summary

Thank You

Questions

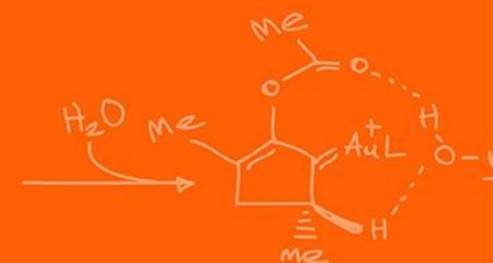
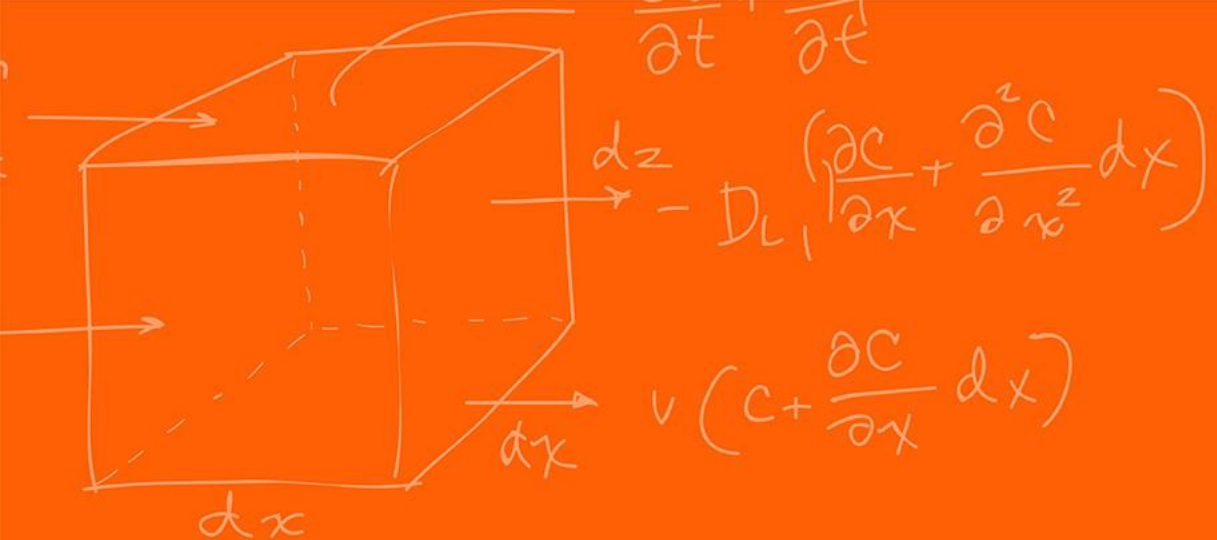
Contact Information

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Questions?





The Grainger College of Engineering

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